Instructions: You will work on these exercises during the practical session in the sixth week of the course. We will cover exercises 1, 2a and 3. We encourage you try to do them before the session.

Exercise 1: Valid Kernels

Which of the following are valid kernels? Justify your answer.

(a) $k(\mathbf{x}, \mathbf{z}) = \|\mathbf{x} - \mathbf{z}\|_2^2$.

Solution:

This is not a valid kernel. A valid kernel must produce a positive semidefinite Gram matrix for every finite set of inputs.
Consider the two points $\mathbf{x}_1 = \mathbf{0}$ and $\mathbf{x}_2 = \mathbf{e}_1$, where $\mathbf{e}_1$ is the first standard basis vector. The Gram matrix is

$$K = \begin{pmatrix} k(\mathbf{0}, \mathbf{0}) & k(\mathbf{0}, \mathbf{e}_1) \\ k(\mathbf{e}_1, \mathbf{0}) & k(\mathbf{e}_1, \mathbf{e}_1) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For the vector $\mathbf{c} = (1, -1)^\top$,

$$\mathbf{c}^\top K \mathbf{c} = -2 < 0.$$

Thus the Gram matrix is not positive semidefinite, so $k(\mathbf{x}, \mathbf{z}) = \|\mathbf{x} - \mathbf{z}\|_2^2$ is not a valid kernel.

(b) $k(\mathbf{x}, \mathbf{z}) = (2\mathbf{x}^\top \mathbf{z} + 1)^3$.

Solution:

This is a valid kernel. The linear kernel $\mathbf{x}^\top \mathbf{z}$ is valid, and multiplying a valid kernel by the positive constant 2 preserves validity. The constant kernel 1 is also valid, and the sum of valid kernels is valid. Finally, raising a valid kernel to the positive integer power 3 preserves validity because this is a product of valid kernels. Therefore

$$k(\mathbf{x}, \mathbf{z}) = (2\mathbf{x}^\top \mathbf{z} + 1)^3$$

is a valid polynomial kernel.

Exercise 2: The Kernel Trick

In the lectures, we have seen how the kernel trick extends linear models to nonlinear ones by replacing scalar products with a kernel function. In this exercise, you will apply this idea to different models.

(a) The nearest-neighbour classifier assigns a new input vector $\mathbf{x}$ to the same class as that of the nearest training example $\mathbf{x}_n$. In the standard formulation, the distance is measured using the squared Euclidean distance

$$\|\mathbf{x} - \mathbf{x}_n\|^2.$$

Express this distance in terms of scalar products and derive the corresponding nearest-neighbour classifier for a general nonlinear kernel.

Solution:

Expanding the squared Euclidean distance gives

$$\|\mathbf{x} - \mathbf{x}_n\|^2 = (\mathbf{x} - \mathbf{x}_n)^\top (\mathbf{x} - \mathbf{x}_n) = \mathbf{x}^\top \mathbf{x} - 2\mathbf{x}_n^\top \mathbf{x} + \mathbf{x}_n^\top \mathbf{x}_n.$$

Suppose the data are mapped to a feature space by a feature map $\phi(\cdot)$ with kernel

$$k(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z}).$$

The squared distance in feature space is therefore

$$\|\phi(\mathbf{x}) - \phi(\mathbf{x}_n)\|^2 = k(\mathbf{x}, \mathbf{x}) + k(\mathbf{x}_n, \mathbf{x}_n) - 2k(\mathbf{x}_n, \mathbf{x}).$$

Hence, the kernel nearest-neighbour classifier predicts the class of the training sample $\mathbf{x}_{n^*}$ such that

$$\begin{aligned} n^* &= \arg\min_n \left[k(\mathbf{x}, \mathbf{x}) + k(\mathbf{x}_n, \mathbf{x}_n) - 2k(\mathbf{x}_n, \mathbf{x}) \right] \\ &= \arg\min_n \left[k(\mathbf{x}_n, \mathbf{x}_n) - 2k(\mathbf{x}_n, \mathbf{x}) \right]. \end{aligned}$$

(Since $k(\mathbf{x}, \mathbf{x})$ is independent of n, it can be omitted when performing the minimization).

(b) Consider the binary perceptron classifier with prediction rule

$$\hat{y} = \text{sign}(\mathbf{w}^\top \mathbf{x}),$$

and update rule

$$\mathbf{w} \leftarrow \mathbf{w} + y_n \mathbf{x}_n$$

whenever a training example $(\mathbf{x}_n, y_n)$ is misclassified.

First, show that after training the weight vector can be written as a linear combination of the vectors $y_n \mathbf{x}_n$. Then, derive the kernelized version of the prediction rule.

Note: assume we intialize $\mathbf{w} = \mathbf{0}$.

Solution:

Initially,

$$\mathbf{w} = \mathbf{0}.$$

Each time example n is misclassified, the update

$$\mathbf{w} \leftarrow \mathbf{w} + y_n \mathbf{x}_n$$

adds another copy of $y_n \mathbf{x}_n$ to the weight vector. If this happens α_n times during training, then

$$\mathbf{w} = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n,$$

where α_n is the number of updates caused by the n-th training example.
Substituting this expression into the prediction function gives

$$\mathbf{w}^\top \mathbf{x} = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n^\top \mathbf{x}.$$

Replacing the scalar products by kernel evaluations we obtain:

$$\hat{y} = \text{sign}\left(\sum_{n=1}^{N} \alpha_n y_n k(\mathbf{x}_n, \mathbf{x}) \right).$$

Exercise 3: Deriving the stationary and KKT conditions.

The purpose of this exercise is to show the reasoning behind the Lagrangian and KKT conditions from form the method for solving inequality-constrained optimization problems introduced in the lectures.

Consider the constrained optimization problem:

$$\max_{\mathbf{x}\in\mathbb{R}^2} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) \geq 0.$$

We aim to find candidate points for a global maximum. The restriction $g(\mathbf{x}) \geq 0$ gives us the **primal feasibility** KKT condition. Restricting the domain to this region gives us two types of candidate points: those on the boundary $g(\mathbf{x}) = 0$ (Case I) and those in the region $g(\mathbf{x}) > 0$ (Case II).

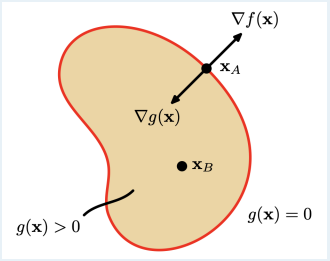
(a) (Case I) If there is a critical point $\mathbf{x}_0$ on the boundary, it means that an unconstrained maximum would occur either on the boundary or outside the feasible region, and the boundary restricts us from reaching that point.

(i) In which direction does $\nabla_{\mathbf{x}} f(\mathbf{x}_0)$ point in this case?

(ii) What about $\nabla_{\mathbf{x}} g(\mathbf{x}_0)$?

(iii) How are the directions of both gradients related? Can you express $\nabla_{\mathbf{x}} f(\mathbf{x}_0)$ in terms of $\nabla_{\mathbf{x}} g(\mathbf{x}_0)$ using a scalar parameter $\mu \geq 0$?

Solution:



(i) $\nabla_{\mathbf{x}} f(\mathbf{x}_0)$ points in the direction of the greatest increase of f from $\mathbf{x}_0$, which is perpendicular to the level curve $g(\mathbf{x}) = 0$, pointing outside the region $g(\mathbf{x}_0) > 0$.

The reason why it is perpendicular to the level curve is as follows: if the direction of maximum increase was not perpendicular to the level curve, then the gradient would have a nonzero component along the curve. This would mean we could move along the curve to increase the value of f, which contradicts the fact that $\mathbf{x}_0$ is a maximum.

(ii) $\nabla_{\mathbf{x}} g(\mathbf{x}_0)$ is perpendicular to the level curve $g(\mathbf{x}) = 0$, pointing towards the region where $g(\mathbf{x}) > 0$.

The fact that it is perpendicular to the level curve follows directly from the definition of a level curve. Using a similar reasoning as before, if the gradient were not perpendicular, one could increase the value of g by moving along the curve, which is impossible since g is constant (0) on the curve.

(iii) The two gradients are anti-parallel, that is:

$$\nabla_{\mathbf{x}} f(\mathbf{x}_0) + \mu \nabla_{\mathbf{x}} g(\mathbf{x}_0) = \mathbf{0}, \quad \mu \geq 0.$$

(b) (Case II) If there is a critical point $\mathbf{x}_0$ inside the feasible region, then the inequality constraint does not restrict the location of the maximum of f.

(i) What is the value of $\nabla_{\mathbf{x}} f(\mathbf{x}_0)$ in this case?

(ii) Can you express this condition using a similar equation as in (a.iii)?

Solution:

(i) $\nabla_{\mathbf{x}} f(\mathbf{x}_0) = \mathbf{0}$.

(ii) This can be written as:

$$\nabla_{\mathbf{x}} f(\mathbf{x}_0) + \mu \nabla_{\mathbf{x}} g(\mathbf{x}_0) = \mathbf{0}, \quad \mu = 0.$$

(c) Combine your results from (a.iii) and (b.ii) into a single condition that all candidate points must satisfy. Congratulations! You have derived the **stationarity condition** for the Lagrangian and the **dual feasibility** KKT condition.

Solution:

$$\nabla_{\mathbf{x}} f(\mathbf{x}_0) + \mu \nabla_{\mathbf{x}} g(\mathbf{x}_0) = \mathbf{0}, \quad \mu \geq 0.$$

(d) Use your findings so far to derive the **complementary slackness** KKT condition.

Solution:

For a candidate point $\mathbf{x}_0$, either it lies on the boundary $g(\mathbf{x}_0) = 0$ (Case I), or it lies inside the feasible region, where $\mu = 0$ (Case II). Therefore, at least one of $g(\mathbf{x}_0)$ and μ must be zero, leading to the **complementary slackness** condition:

$$\mu \, g(\mathbf{x}_0) = 0.$$

(e) How would we define the Lagrangian for the following optimization problems? *Hint: Think about whether $\nabla_{\mathbf{x}} f$ and $\nabla_{\mathbf{x}} g$ would be paralell or anti-paralell in each case.*

(i) $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \geq 0$.

(ii) $\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq 0$.

(iii) $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq 0$.

Solution:

(i) $\mathcal{L}(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu g(\mathbf{x})$ (we are now interested in the direction of maximum **decrease** of f, which is $-\nabla_{\mathbf{x}} f$, which is anti-paralell to $\nabla_{\mathbf{x}} g$).

(ii) $\mathcal{L}(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu g(\mathbf{x})$ (now $\nabla_{\mathbf{x}} g$ points in the opposite direction as in the original problem).

(iii) $\mathcal{L}(\mathbf{x}, \mu) = f(\mathbf{x}) + \mu g(\mathbf{x})$ ($\nabla_{\mathbf{x}} g$ points in the opposite direction as in the original problem, but we consider $-\nabla_{\mathbf{x}} f$, so the gradients are still anti-paralell).

($\mu \geq 0$ in all cases).

Exercise 4: Inequality-constrained optimization

Consider the following 1D optimization problem:

$$\min_{x \in \mathbb{R}} \frac{1}{2}(x-2)^2 \quad \text{subject to} \quad x \geq 4.$$

Of course, this is a very simple problem that can be solved just by evaluating the function in the critical point of the parabola ($x = 2$) and in the extreme of the domain ($x = 4$). However, we will take the long way around to further demystify the method of Lagrange multipliers.

(a) Formulate the Lagrangian of the problem using a Lagrange multiplier $\mu \geq 0$ for the inequality constraint.

Solution:

The Lagrangian is:

$$\mathcal{L}(x, \mu) = \frac{1}{2}(x-2)^2 - \mu(x-4), \quad \mu \geq 0.$$

Note that, as this is a minimization problem, the term with the Lagrangian multiplier is negative, as $-\nabla_{\mathbf{x}} f$ and $\nabla_{\mathbf{x}} g$ will be parallel (pointing in the same direction) at the potential minimums located in the boundary.

(b) Derive the stationarity condition and express the primal variable x in terms of the dual variable μ.

Solution:

The derivative of the Lagrangian w.r.t. x is

$$\frac{\partial \mathcal{L}}{\partial x} = x - 2 - \mu.$$

Setting it to zero, we obtain the stationary condition

$$x = 2 + \mu.$$

(c) Write down the complementary slackness condition.

Solution:

$$\mu(x-4) = 0.$$

(d) Construct the Dual Lagrangian $\tilde{\mathcal{L}}(\mu) = \min_x \mathcal{L}(x, \mu)$ and formulate the dual problem.

Solution:

We substitute the stationary condition $x = 2 + \mu$ into the Lagrangian:

$$\tilde{\mathcal{L}}(\mu) = \frac{1}{2}(\mu)^2 - \mu(\mu - 2) = -\frac{1}{2}\mu^2 + 2\mu.$$

The dual problem is

$$\max_{\mu \geq 0} \tilde{\mathcal{L}}(\mu).$$

(e) Solve the dual problem to find μ^* and then compute the corresponding primal optimum x^*. Is the value located in the boundary or inside the constrained region?

Solution:

$$\frac{d\tilde{\mathcal{L}}}{d\mu} = -\mu + 2; \quad \frac{d\tilde{\mathcal{L}}}{d\mu} = 0 \Leftrightarrow \mu = 2,$$

Since the Dual Lagrangian is concave, the dual maximum occurs at $\mu^* = 2$.
Substituting μ^* in the stationary condition, we obtain

$$x^* = 2 + \mu^* = 4.$$

The optimal value is in the boundary $x - 4 = 0$.

(f) Verify that the obtained solution meets the KKT conditions.

Solution:

- Primal feasibility: $x^* - 4 = 0 \geq 0$.
- Dual feasibility: $\mu^* = 2 \geq 0$.
- Complementary slackness: $\mu^*(x^* - 4) = 2(4 - 4) = 0$.